Reducing Pronoun Accessibility to Presupposition Satisfaction

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Introduction

The meaning of certain sentences maps neatly to classical logic.

(1) a. There is a phone-book.

b. $\exists x, \textbf{phone-book}(x) \wedge \textbf{there}(x)$

(2) a. A dog chased its tail.

b. $\exists x, \textbf{dog}(x) \wedge \textbf{chased}(\textbf{tail}(x))(x)$

But pronouns referring to indefinites are more liberal than the variables of classical logic:

(3) a. There is a phone-book and it is in the cabinet.

b. $(\exists x, \textbf{phonebook}(x) \wedge \textbf{there}(x)) \wedge (\textbf{cabinet}(x))$

Accessibility conditions for pronouns

When can a pronoun refer to a given (indefinite) antecedent?

A number of sentences are surprising from the perspective of classical logic:

(4) a. There is a phone-book and it is in the cabinet.
(cross-conjunction)

b. If there is a phone-book, it is in the cabinet.
(donkey conditional)

c. Either there isn't a phone-book or it is in the cabinet.
(bathroom sentences)

d. There might be a phone-book. It would be in the cabinet.
(modal subordination)

But not everything goes:

(5) a. # There isn't a phone-book and it is in the cabinet.

b. # If there is a phone-book, all is good. It is in the cabinet.

c. # Either there is a phone-book or it is in the cabinet.

d. # There might be a phone-book. It is in the cabinet.

Old Intuition: pronoun accessibility is connected to presupposition satisfaction.

[Heim, 1982, Kamp et al., 2011, among many others]

(6) a. There is a phone-book and it is in the cabinet.

b. If there is a phone-book, it is in the cabinet.

c. There might be a phone-book. It would be in the cabinet.

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b. If there is a phone-book, it is in the cabinet.

c. There might be a phone-book. It would be in the cabinet.

(7) a. # There isn't a phone-book and it is in the cabinet.

b. # If there is a phone-book, all is good. It is in the cabinet.

c. # Either there is a phone-book or it is in cabinet.

You can only refer to the phone-book if the existence of the phone-book can be presupposed when the pronoun is used?

Parallels pronouns/presuppositions

(8) a. There was a phone-book and it was in the cabinet.

b. There was a phone-book and there still is.

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(12) a. # There wasn't a phone-book and it is in the cabinet.

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b.# There might have been a phone-book and there still is.

(14) a.# Either there was a phone-book or it was in the cabinet.

b.# Either there was a phone-book or there still is.

Accessibility Generalization

In a context "…*a/some* RESTR SCOPE …*it* …", *it* may co-refer with *a* RESTR if and only if the presupposition that RESTR $\cap$ SCOPE $\neq \varnothing$ is satisfied at the point where the pronoun is used.

(Cf also [Mandelkern and Rothschild, 2019])

Accessibility conditions for pronouns

When can a pronoun refer to a given antecedent?

Reduces to:

Presupposition satisfaction for presuppositions

When is a presupposition met?

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- But they do not reduce pronoun accessibility to presupposition satisfaction.
- As a result, they fall short of validating the generalization in full.

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- Present arguments (old and new) for the accessibility generalization.
- Propose a system in which pronouns are interpretable as soon as the existence presupposition is met.
- Pronouns pick a referent from a predicate via a choice function.
- It has broad empirical coverage.
- It does not require significantly altering the semantics.

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Remarks:

Accessibility Generalization

In a context "…*a/some* RESTR SCOPE …*it* …", *it* may co-refer with *a* RESTR if and only if the presupposition that RESTR $\cap$ SCOPE $\neq \varnothing$ is satisfied at the point where the pronoun is used.

(15) ?Jane is married but I've only met them once.

Remarks:

- The generalization requires a DP antecedent.

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In a context "…*a/some* RESTR SCOPE …*it* …", *it* may co-refer with *a* RESTR if and only if the presupposition that RESTR $\cap$ SCOPE $\neq \varnothing$ is satisfied at the point where the pronoun is used.

Remarks:

- The generalization requires a DP antecedent.
- Typically, all theories validate:
  pronoun accessible $\Rightarrow$ existence presupposition met
- The reverse is more contentious and typically not validated:
  existence presupposition met $\Rightarrow$ pronoun accessible

- The generalization is motivated through case studies.
- Each show a correlation between a certain pattern of presupposition satisfaction and pattern of pronoun accessibility.
- We will see one exception to the generalization but it is a principled one.

Case study I: anaphora in conjunction

(16) There is a phone-book$_i$ and $\underbrace{\text{it}_i \text{ is in the cabinet}}_{\text{pres: there is a phone-book}}$.

(16) There is a phone-book$_i$ and it$_i$ is in the cabinet.

pres: $\varnothing$

From theories of presupposition satisfaction:
p and $\underline{p}q$ does not carry a presupposition.

(17)# $\underline{\text{It}_i \text{ is in the cabinet}}$ and there is a phone-book$_i$.

pres: there is a phone-book

From theories of presupposition satisfaction:
The presupposition of "$\underline{p'}p$ *and* q" is p'
[Schlenker, 2009, Fox, 2013, George, 2008].

Case study II: cataphora in conjunction

(17) # $\underline{\text{It}_i \text{ is in the cabinet and there is a phone-book}_i}$.

pres: there is a phone-book

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The presupposition of "$\underline{p'}p$ *and* q" is p'
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Case study II: cataphora in conjunction

The ban on cataphora is related to the impossibility of right-to-left presupposition satisfaction [Mandelkern et al., 2020].

(18) a. There is a phone-book$_i$ and it$_i$ is in the cabinet.

b. There was a phone-book and there still is a phone-book.

(19) a. # It$_i$ is in the cabinet and there is a phone-book$_i$.

b. # There still is a phone-book and there was a phone-book.

Case study III: exceptions to the ban on cataphora

But there are exceptions to the ban on right-to-left presupposition satisfaction:

(20) Jane's no longer in Norway but she was there at some point.

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(20) Jane's no longer in Norway but she was there at some point.

Such configurations also license cataphora:

(21) She$_i$ isn't in Norway now but one of my students$_i$ studied there.

NB: *but* also allows left-to-right presupposition satisfaction and regular anaphora

(22) Jane was in Norway at some point but she's no longer there.

(23) One of my students studied in Norway but she isn't there now.

Case study IV: Bathroom sentences

(24) Either there wasn't a phone-book$_i$
or $\underline{\text{it}_i \text{ was in the cabinet}}$.
pres.: there was a phone-book

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or $\underline{\text{it}_i \text{ was in the cabinet}}$.
pres.: there was a phone-book

From theories of presupposition satisfaction:
either not p or $\underline{p}q$ does not carry a presupposition.

Traditional Dynamic Semantics doesn't predict this. Treating pronoun accessibility and presuppositions does not *ipso facto* mean guaranteeing the accessibility generalization.

Case study V: accessibility via pragmatic inferences

(25) Bill believes that I don't have a sister$_{¡}$.
$\rightsquigarrow$ *I have a sister.*

Maximize Presupposition: [Heim, 1991, Chemla, 2008]
When speaker is likely well-informed about *p*, "*Bill believes p*" conveys *p* is false.

Case study V: accessibility via pragmatic inferences

A presupposition can be satisfied by such a pragmatic inference:

(26) Bill believes that I don't have a sister$_i$.
Yet, he's met my sister$_i$ several times.

Case study V: accessibility via pragmatic inferences

And a pronoun can likewise be made available:

(27) Bill believes I don't have a sister$_i$.
Yet, he's met her$_i$ several times.

Case study V: accessibility via pragmatic inferences

The inference is defeasible; when defeated, the pronoun becomes unavailable:

(28) Bill believes, <u>and he's correct,</u> that I don't have a sister$_i$.
Yet, he's met her$_i$ several times.

Case study VI : cataphora and condition C

The following example looks problematic (A. Anvari, p.c.)

(29) They$_i$ know that there is someone$_i$ in the lobby.

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- The sentence as a whole already presupposes that there is someone in the lobby.
- This happens to be the licensing condition on the pronoun.
- No reason to assume this condition cannot be met.

Case study VI : cataphora and condition C

The following example looks problematic (A. Anvari, p.c.)

(29) They$_i$ know that there is someone$_i$ in the lobby.
$\rightsquigarrow$ *# a certain person in the lobby knows that there is someone in the lobby.*

- The sentence as a whole already presupposes that there is someone in the lobby.
- This happens to be the licensing condition on the pronoun.
- No reason to assume this condition cannot be met.

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$\rightsquigarrow$ *independent need for condition C.*

Outside of condition C:

(30) ✓ The person that placed it_i there knows that there is a phone-book_i in the cabinet.

Interim summary

- **Accessibility generalization:** in the presence of an indefinite antecedent, a pronoun is accessible iff a witness can be presuppposed to exist.
- Exception: condition C.
- Covers:
  - Simple conjunctions.
  - Ban on cataphora.
  - Exceptions to the ban.
  - Bathroom sentences.
  - Intrusion of pragmatic inferences in pronoun's accessibility conditions.
  - Cataphora in condition C.

Proposal

Pre-emptive summary

- Pronouns pick a referent from a certain *witness set* via a choice function.
- Pronouns naturally presuppose the witness set is not empty.
- As soon as this presupposition is met, the pronoun receives an interpretation.
- We can assume any theory of presupposition satisfaction, including "*explanatory*" theories.
  [Schlenker, 2009, Fox, 2013, George, 2008, Rothschild, 2011].

Pronoun structure

G and f are two fixed parameters of interpretation:

- for every i and w, G_{iw} is a set called the witness set.
  (e.g. the set of phone-books)
- for every i and w, f_{iw} is a choice function:
  $\forall S, f(S) \in S$
  (e.g. the phone-book we want to pick)

(31) $[\![\text{it}_i]\!]^w = [\underline{G_{iw} \neq \varnothing}]\, f_{iw}(G_{iw})$

Pronoun structure

There are principles constraining G_{iw}. For now, we assume:

Expressivity Constraint

For any property P, there is an i such that $\forall w, G_{iw} = P(w)$

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Indexing Constraint (provisional)

If there is a constituent "*[a NP]*$_i$ *VP*" in the neighboring linguistic context, then it has to be the case that $G_{iw} = [\![\text{NP}]\!]^w \cap [\![\text{VP}]\!]^w$ for all w.

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(32) There is a phone-book$_i$ …
$\rightsquigarrow G_{iw} = [\![\text{phone-book}]\!]^w \cap [\![\text{there}]\!]^w$

Provisional rule for f_{iw}

No constraint apart from it being a choice function.

Anaphora in conjunction

(33) a. There is a phone-book$_i$ and it$_i$ is in the cabinet.

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(33) a. There is a phone-book$_i$ and it$_i$ is in the cabinet.

b. $[\![\text{it}_i]\!] = \dfrac{[[\![\text{phone-book}]\!]^w \cap [\![\text{there}]\!]^w \neq \varnothing]}{f_{iw}([\![\text{phone-book}]\!]^w \cap [\![\text{there}]\!]^w)}$

c. **presupposes:** there is a phone-book
denotes: a "*certain*" phone-book that there is

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d. *There is a phone-book and a certain phone-book that there is is in the cabinet.*

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(34) a. If there is a phone-book$_i$, it$_i$ is in the cabinet.

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c. **presupposes:** there is a phone-book
denotes: a "*certain*" phone-book that there is

d. *If there is a phone-book, a certain phone-book that there is is in the cabinet.*

What about *donkey sentences*?

Quantificational cases

(35) Every farmer owns a donkey$_i$.
Few of them feed it$_i$ oats.

(36) Every farmer who owns a donkey$_i$ feeds it$_i$ oats.

What does the indexing constraint say about embedded indefinites?

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Indexing constraint

If there is a constituent "*[a NP]$_i$ VP*" in the context, then $G_{iw} = [\![\text{NP}]\!]^w \cap [\![\text{VP}]\!]^w$ for all w.

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If there is a constituent "*[a NP]$_i$ VP*" in the context, then $G_{iw}(g) = [\![\text{NP}]\!]^{w,g} \cap [\![\text{VP}]\!]^{w,g}$ for all w.

In this case: $G_{iw}(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey in } w \\ g(x) \text{ owns } y \text{ in } w \end{array} \right\}$

This requires a minimal adaptation of the pronouns interpretation:

(38) $[\![\text{it}_\text{i}]\!]^{w,g} = f_{iw}(G_{iw}(g))$

Sore point: in-scope bound pronouns have to be treated differently!

$$[\![\text{it}_i]\!]^{w,g} = g(i)$$

(39) A dog chased its tail.

$\rightsquigarrow$ *needs fixing!*

(40) a. Every farmer owns a donkey$_i$.

b. $G_{iw}(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey in } w \\ g(x) \text{ owns } y \text{ in } w \end{array} \right\}$

(41) a. Few of them feed it$_i$ oats.

b. Few of them $\lambda x.\ x$ feed it$_i$ oats.

(40) a. Every farmer owns a donkey$_i$.

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pres.: $G_{iw}(g) \neq \varnothing$

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b. Few of them $\lambda x.$ $\underline{x \text{ feed it}_i \text{ oats}}$

pres.: $g(x)$ owns a donkey in w

Universal presupposition projection out of *few*:

(42) a. Few of the farmers still have a donkey.
$\rightsquigarrow$ *all of them had a donkey*

b. Few of the farmers $\lambda x.$ $\underbrace{x \text{ still have a donkey}}_{\text{pres.: } g(x) \text{ had a donkey}}$.

c. $\underbrace{\text{Few of the farmers } \lambda x.\ x \text{ still have a donkey}}_{\text{pres.: all farmers had a donkey}}$.

(43) a. Every farmer owns a donkey$_i$.
Few of them feed it$_i$ oats.
≈ *few of them feed a certain donkey they own oats.*

b. Few of them $\lambda x.$ $\underbrace{x \text{ feed it}_i \text{ oats}}_{\text{pres.: } g(x) \text{ owns a donkey in } w}$

c. $\underbrace{\text{Few of them } \lambda x.\ x \text{ feed it}_i \text{ oats}}_{\text{pres.: all of them own a donkey in } w}$

The same reasoning applies to the donkey sentence.

(44) a. Every farmer who owns a donkey$_i$ feeds it$_i$ oats.

b. Every farmer who owns a donkey$_i$

$$\lambda x.\ \underbrace{x \text{ feeds it}_i \text{ oats}}_{\text{pres.: } G_{iw}(g) \neq \varnothing}$$

c. $$G_{iw}(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey in } w \\ g(x) \text{ owns } y \text{ in } w \end{array} \right\}$$

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c. $G_{iw}(g) = \left\{ y \;\middle|\; \begin{array}{l} y \text{ is a donkey in } w \\ g(x) \text{ owns } y \text{ in } w \end{array} \right\}$

d. $\approx$ *every farmer who owns a donkey feeds a certain donkey they own oats.*

Takeaway: quantificational sentences require recognizing that the set G_{iw} may depend on the value of certain binders.

$\exists$ AND $\forall$ READINGS OF PRONOUNS

Pronoun structure

(45) $[\![\text{it}_\text{i}]\!]^{w,g} = f_{iw}(G_{iw}(g))$

If there is just one witness, different choice functions will yield the same truth-conditions: $f(\{p\}) = p$

(46) There is a phone-book$_i$ and it$_i$ is in the cabinet.

What if there are multiple phone-books?

(47) There is a phone-book$_i$ and it$_i$ is in the cabinet.

Empically, is the sentence true if:

- there are two phone-books
- one is in the cabinet, the other isn't ?

(48) There is a phone-book$_i$ and it$_i$ is in the cabinet.

Two hypothetical readings have been discussed:

- $\forall$-reading : *every phone-book that there is is in the cabinet*
- $\exists$-reading : *at least one phone-book is in the cabinet*

Which readings obtain?

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Two hypothetical readings have been discussed:

- $\forall$-reading : *every phone-book that there is is in the cabinet*
- $\exists$-reading : *at least one phone-book is in the cabinet*

Which readings obtain? $\rightsquigarrow$ *may depend on the antecedent-pronoun configuration*

Although typically discussed for donkey sentences [Kanazawa, 1994], the question is raised in *all* environments.

(49) a. If there is a phone-book, it is in the cabinet.

b. Either there isn't a phone-book or it's in the cabinet.

c. Every person that has a phone-book puts it in their cabinet.

What we know about the readings of anaphora

The presence of a default reading has been confirmed experimentally many times in donkey sentences. [Foppolo, 2008, Denić and Sudo, 2022, Sun et al., 2020]

(50) a. Some farmer who has a donkey feeds it oats.
$\rightsquigarrow$ *feeds one of their donkeys oats* ($\exists$, 90%)

b. Every farmer who has a donkey feeds it oats.
$\rightsquigarrow$ *feeds all of their donkeys oats* ($\forall$, 60%)

c. No person who has a donkey feeds it oats.
$\rightsquigarrow$ *feeds any of their donkeys oats* ($\exists$, 90%)

What we know about the readings of anaphora

But certain lexical items are biased and can override the "*default*" reading.

(51) a. No person who had an umbrella left it at home today.
$\rightsquigarrow$ *left all of their umbellas at home* ($\forall$)

b. Jane has an umbrella and she left it at home.
$\rightsquigarrow$ *left all of their umbellas at home* ($\forall$)
[Chatain, 2018, Mandelkern, 2020]

There remain some knowledge gaps:

- What about other environments (*and*, bathroom sentences)?
- What about intuitions of uniqueness?
- What pragmatic factors determines the reading?
- What "*scope*" does this pseudo-quantification get? [Brasoveanu, 2007, Chatain, 2018, Chierchia, 2022]

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$\rightsquigarrow$ *joint work with Benjamin Spector and Nina Gregorio*

A vexed question:

- [Mandelkern, 2020]: *"This is a complicated issue, which I won't explore in detail here."*
- [Elliott, 2020] : *"How to capture existential vs. universal readings is a thorny issue, and [...] therefore we leave a more thorough exploration of donkey anaphora [...] to future work."*

$\rightsquigarrow$ likewise, my answer will be incomplete!

Generalizing to the worst case, I try to derive all possible readings, not accounting for "*default readings*".

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Choice of f

A sentence S is:

- definitely true if true for all choices of f
- definitely false if false for all choices of f
- undetermined otherwise

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A sentence S is:

- definitely true if true for all choices of f
- definitely false if false for all choices of f
- undetermined otherwise

Depending on the QUD and other pragmatic factors, undetermined may resolve to true or false [Kriz, 2015, Champollion et al., 2017].

(52) Every farmer who has a donkey$_i$ feeds it$_i$ oats.
where $G_{iw}(g)$ = the set of donkeys owned by $g(x)$ in w

a. **Definitely true:**
for all f,
every farmer who has a donkey
feeds f(donkey of $g(x)$) oats

b. **Definitely false:**
for no f,
every farmer who has a donkey
feeds f(donkey of $g(x)$) oats

This determines the following truth-/falsity-conditions.

(53) Every farmer who has a donkey$_i$ feeds it$_i$ oats.

a. every farmer feeds all of them oats (true, $\forall$)

b. some farmer feeds none of them oats (false, $\neg\exists$)

c. some farmer feeds some but not all of them oats (undetermined)

Depending on the pragmatics, the reading can oscillate between $\exists$ and $\forall$.

Positive: we predict $\exists$ and $\forall$ readings for all sentence types.

Positive: we predict $\exists$ and $\forall$ readings for all sentence types.
Negative: we don't predict or expect there to be default readings, nor their preponderance.

Previous works, precedents, inspirations

Dynamic Semantics only validates one direction of the accessibility generalization: when *it* is used, the existence of the phone-book can be presupposed.

(54) Predicted #

a. Either there wasn't a phone-book or it was in the cabinet. (bathroom sentences)

b. It's not true that there isn't a phone-book. It's hidden in the cabinet. (double negation)

Previous work

Fixes [Elliott, 2020, Krahmer and Muskens, 1995] exist, typically at two costs:

- …richer spaces of meaning.
- …leave other empirical gaps open [Hofmann, 2019].

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- …richer spaces of meaning.
- …leave other empirical gaps open [Hofmann, 2019].

(55) The claim that there is a phone-book in this house is surprising.
If it is true, where would one find it?

(56) The claim that there is a phone-book in this house is of course false.
If this claim had been true, it would have been in the cabinet.

Can Dynamic Semantics ever validate the accessibility generalization?
$\rightsquigarrow$ *perhaps!*

This theory finds parallels in:

1. [Hofmann, 2019]'s ICDRT
2. E-type approaches

Precedents and inspirations: ICDRT

[Hofmann, 2019]'s ICDRT:

- Indefinites introduce individual concepts in a dynamic system, which carry existence presuppositions.
- Pronouns' accessibility conditions are duplicated: they are dictated by both dynamic meanings and presupposition satisfaction.

Precedents and inspirations: E-type approaches

E-type theories assume:

- Pronouns are definite descriptions:
  it = the phone-book that there was.
- Fregean treatment of definite descriptions
  (existence **+ uniqueness** presuppositions).
- Accessibility conditions reduced to presupposition satisfaction.

[Evans, 1977, Geach, 1964, Parsons, 1978, Neale, 1990, Heim, 1990, Büring, 2004, Elbourne, 2005, Elbourne, 2001]

Precedents and inspirations: E-type approaches

Two long-standing issues:

- Formal link: which description is recovered?

(57) a. Exactly nine of the ten marbles have been found.
We're still looking for it.

b. Exactly one of the ten marbles is missing.
We're still looking for it.

Precedents and inspirations: E-type approaches

Two long-standing issues:

- Formal link: which description is recovered?
- Pronouns can be interpreted even when no uniqueness inferences can be derived.

(57) (Since sage plants are sold in packs of 8,)
Every client who bought a sage plant has bought seven others along with it. [Heim, 1982]

Precedents and inspirations: E-type approaches

By contrast, in this theory:

- Formal link: there is formal co-indexation.
- No commitment to uniqueness (through choice functions).
- We don't equate pronouns with definites.

The idea of using choice functions in E-type theories is not new. [Heusinger, 2004, Egli and Von Heusinger, 1995, van der Does, 1993]

The idea of using choice functions in E-type theories is not new. [Heusinger, 2004, Egli and Von Heusinger, 1995, van der Does, 1993]

Original to my work:

- an explicit theory of the formal link,
- connection to presupposition satisfaction made explicit,

Conclusion

Summary

- With the proposed theory, pronoun accessibility fully reduce to presupposition satisfaction.

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- With the proposed theory, pronoun accessibility fully reduce to presupposition satisfaction.
- No need for enrichment ; the semantics is kept mostly intact.
- **Predictions:**
  - Bathroom sentences.
  - Donkey, modal subordination.
  - Availability and unavailability of cataphora.
  - Intrusion of pragmatic inferences in pronoun accessibility.

Extensions

- Developing to all types of antecedent (account for differences)
- Stone disjunction (cf (58)) and restrictions on their use.
- Paycheck pronouns and restrictions on their use.

(58) If Jane buys a donkey or a goat, she'll take good care of it.

Thank you!

Brasoveanu, A. (2007).
Structured Nominal and Modal Reference.
PhD thesis, Rutgers University New Brunswick, NJ.

Büring, D. (2004).
Crossover situations.
Natural Language Semantics, 12(1):23–62.

Champollion, L., Bumford, D., and Henderson, R. (2017).
Homogeneity in donkey anaphora.
(2016):1–38.

Chatain, K. (2018).
Gaps in the interpretation of pronouns.
In *Semantics and Linguistic Theory*, volume 28, pages 177–196.

Chemla, E. (2008).
An epistemic step for anti-presuppositions.
Journal of Semantics, 25(2):141–173.

Chierchia, G. (2022).
'People are fed up; don't mess with them.'Non-quantificational Arguments and Polarity Reversals.
Journal of Semantics, 39(3):475–521.

Denić, M. and Sudo, Y. (2022).
Donkey anaphora in non-monotonic environments.
Journal of Semantics, 39(3):443–474.

Egli, U. and Von Heusinger, K. (1995).
The epsilon operator and E-type pronouns.
Amsterdam Studies in the Theory and History of Linguistic Science Series 4, pages 121–121.

Elbourne, P. (2001).
E-type anaphora as NP-deletion.
Natural Language Semantics, 9(3):241–288.

Elbourne, P. (2005).
***Situations and Individuals*, volume 90.**
Mit Press Cambridge, MA.

Elliott, P. D. (2020).
Towards a principled logic of anaphora.

Evans, G. (1977).
Pronouns, quantifiers, and relative clauses (I).
Canadian journal of philosophy, 7(3):467–536.

Foppolo, F. (2008).
The puzzle of donkey anaphora resolution.
In Schardl, A., Walkow, M., and Abdurrahman, M., editors, *North East Linguistics Society (NELS)*, volume 38, pages 297–310. GLSA.

Fox, D. (2013).
Presupposition projection from quantificational sentences: Trivalence, local accommodation, and presupposition strengthening.
In Caponigro, I. and Cecchetto, C., editors, *From Grammar to Meaning*, pages 201–232. Cambridge University Press.

Geach, P. T. (1964).
Reference and Generality.
Cornell University Press.

George, B. (2008).
Predicting presupposition projection: Some alternatives in the strong Kleene tradition.
Manuscript, UCLA.

Heim, I. (1982).
The Semantics of Definite and Indefinite Noun Phrases.
PhD thesis.

Heim, I. (1990).
E-Type pronouns and donkey anaphora.
Linguistics and Philosophy, 13(2):137–177.

Heim, I. (1991).
Artikel und definitheit.
Semantik: ein internationales Handbuch der zeitgenössischen Forschung, pages 487–535.

Heusinger, K. V. (2004).
Choice Functions and the Anaphoric Semantics of Definite NPs.
Research on Language and Computation, 2(3):309–329.

Hofmann, L. (2019).
The anaphoric potential of indefinites under negation and disjunction.
In *Proceedings of the 22nd Amsterdam Colloquium*, pages 181–190.

Kamp, H., Van Genabith, J., and Reyle, U. (2011).
Discourse representation theory.
In *Handbook of Philosophical Logic*, pages 125–394. Springer.

Kanazawa, M. (1994).
Weak vs. strong readings of donkey sentences and monotonicity inference in a dynamic setting.
Linguistics and Philosophy, 17(2):109–158.

Krahmer, E. and Muskens, R. (1995).
Negation and disjunction in discourse representation theory.
Journal of Semantics, 12(4):357–376.

Kriz, M. (2015).
Aspects of Homogeneity in the Semantics of Natural Language.
PhD thesis, University of Vienna.

Mandelkern, M. (2020).
Witnesses.

Mandelkern, M. and Rothschild, D. (2019).
Definiteness projection.
Natural Language Semantics, pages 1–33.

Mandelkern, M., Zehr, J., Romoli, J., and Schwarz, F. (2020).
We've discovered that projection across conjunction is asymmetric (and it is!).
Linguistics and Philosophy, 43:473–514.

Neale, S. (1990).
Descriptions.

Parsons, T. (1978).
Pronouns as paraphrases.

Rothschild, D. (2011).
Explaining presupposition projection with dynamic semantics.
Semantics and Pragmatics, 4(3):1–43.

Schlenker, P. (2009).
Local contexts.
Semantics and pragmatics, 2:1–3.

Sun, C., Rothschild, D., and Breheny, R. (2020).
Exploring the existential/universal ambiguity in singular donkey sentences.
In *Proceedings of Sinn Und Bedeutung*, volume 24, pages 289–305. PKP.

van der Does, J. (1993).
The dynamics of sophisticated laziness.
Plurals and Anaphora. Dyana-2 Deliv, pages 1–52.

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